

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260520_202759_i
- started_at_local: 2026-05-20 20:27:59
- finished_at_local: 2026-05-20 20:54:46
- duration_hhmm: 00:27
- duration_seconds: 1606.713
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Run time (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1513.481557	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.140178	0.067918	3828.667229	0.354609	0.104682	0.117759	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1502.802957	0.07226	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1489.335586	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.230747	0.158487	296.350486	0.230747	0.073705	0.131563	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.20283	0.13057	347.631771	0.20283	0.023657	0.089668	0.073958	0.62	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.214732	0.142472	198.764786	0.214732	0.046459	0.108455	0.488141	1.7	False
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	101.700771	0.213387	0.150269	0.173523	0.837837	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1513.481557 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.140178, selector regret 0.067918, runtime-adjusted gap 0.354609.
- Family transfer 0.104682 and combined transfer 0.117759.
- Runtime 3828.667229 ms, runtime inflation 1.529709, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1502.802957 ms, runtime inflation -0.007056, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1489.335586 ms, runtime inflation -0.015954, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.230747, selector regret 0.158487, runtime-adjusted gap 0.230747.
- Family transfer 0.073705 and combined transfer 0.131563.
- Runtime 296.350486 ms, runtime inflation -0.804193, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:limited_three_opt:edge_preserving_restart:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.004, "candidate_limit_offset": 3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "annealed_multi_start", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-conditioned selector from frozen portfolio: structure-first heuristic mapping with bounded schedule tuning.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "det_tsp_hh_oracle_style_portfolio_epoch1", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.20283, selector regret 0.13057, runtime-adjusted gap 0.20283.
- Family transfer 0.023657 and combined transfer 0.089668.
- Runtime 347.631771 ms, runtime inflation -0.77031, mean novelty 0.073958, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed_multi_start:cheapest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag4:cand3:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": 3, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "candidate_pruned_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Interpretable, instance-adaptive scheduler selecting from a frozen TSP heuristic portfolio with bounded online adjustments via stagnation, improvement, and failed-perturbation triggers.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.1, "name": "deterministic_instance_adaptive_portfolio_v2", "nearest_neighbor_trap_heuristic":

"candidate_pruned_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2,
"stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 4, "temperature_scale": 1.25,
"time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt")

phase8_ilm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.214732, selector regret 0.142472, runtime-adjusted gap 0.214732.
- Family transfer 0.046459 and combined transfer 0.108455.
- Runtime 198.764786 ms, runtime inflation -0.86867, mean novelty 0.488141, and complexity 1.7.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:limited_three_opt:edge_preserving_restart:stag4:cand-1:restart1.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -1, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic interpretable instance-adaptive hyper-heuristic controller selecting among a frozen TSP heuristic portfolio with bounded online tuning from instance descriptors and runtime stagnation signals.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.11, "name": "tsp_hh_portfolio_v2_deterministic", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "policy": {"deterministic": true, "deterministic_tuning_rules": {"annealing_acceptance": {"acceptance_bias_rule": "acceptance_bias + 0.008*(stagnation length - 1)", "acceptance_clamp": [-0.01, 0.02], "temperature_clamp": [0.5, 1.8], "temperature_scale_rule": "temperature_scale * (1.0 + 0.45*(distance_cv - 0.5))", "candidate_limit": {"clamp": [-6, 6], "offset_formula": "candidate_limit_offset + (1 if grid_likeness >= 0.40 else -1) + (-1 if recent improvement rate <= low_improvement_threshold else 0)", "restart_intensity": {"clamp": [-3, 4], "int_formula": "restart_offset + (1 if stagnation length >= stagnation_threshold else 0) + (1 if failed perturbation count >= failed_perturbation_threshold else 0) - (1 if recent improvement rate > low_improvement_threshold else 0)", "diversity_guard_for_diversity_failure_mode": {"condition": [{"non-improving streak", ">=", 2}], "rule": "if recent accepted swaps are low and non-improving streak >= 2, run random_like_heuristic once at the next stagnation checkpoint instead of stagnation_switch_heuristic"}, "fallback": "default_heuristic", "instance_selection": [{"if": [{"bottleneck / two-cluster score", ">=", 0.58}, {"distance_cv", ">=", 0.45}], "then": "two_cluster_bottleneck_heuristic"}, {"if": [{"corridor score / elongated structure", ">=", 0.46}], "then": "corridor_heuristic"}, {"if": [{"grid-likeness", ">=", 0.4}], "then": "grid_like_heuristic"}, {"if": [{"clustering score", ">=", 0.52}], "then": "clustered_heuristic"}, {"if": [{"nearest-neighbor trap score", ">=", 0.26}], "then": "nearest_neighbor_trap_heuristic"}], "online_schedule": [{"before_trigger": {"acceptance_bias_effect": ["instance distance_std_norm", "affine", -0.002, 0.01, "clip", [-0.01, 0.02]], "candidate_limit_offset_effect": ["instance grid-likeness", "lin", -1.5, 1.5, "clip", [-6, 6]], "restart_intensity_effect": ["instance bottleneck / two-cluster score", "lin", -1, 2, "clip", [-3, 4]], "temperature_scale_effect": ["instance distance_cv", "affine_to_scale", 0.95, 1.15, "clip", [0.5, 1.8]]}, "phase_split_by": "time_budget_trigger", {"if": [{"stagnation length", ">=", "stagnation_threshold"]}, "then_switch_heuristic": "stagnation_switch_heuristic", "trigger": "stagnation length", {"if": [{"recent improvement rate", "<=", "low_improvement_threshold"]}, "then": {"candidate_list_action": "tighten candidate list deterministically by 1 step (or apply candidate_limit_offset if late-phase)", "failed_perturbation_count_action": {"do": "increase restart intensity deterministically by 1 step", "if_failed_count": ["failed perturbation count", ">=", "failed_perturbation_threshold"]}, "trigger": "stagnation / low improvement", {"if": [{"time budget used", ">=", "time_budget_trigger"]}, "then": {"candidate_list_action": "set candidate limit using candidate_limit_offset and grid-likeness sign", "restart_intensity_action": "reduce perturbation aggressiveness by 1 step if recent improvement rate > low_improvement_threshold else keep", "trigger": "late phase tightening"}], "priority_order": ["two_cluster_bottleneck", "corridor", "grid_like", "clustered", "nearest_neighbor_trap", "default"]}, "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 4, "temperature_scale": 1.15, "time_budget_trigger": 0.62, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 101.700771 ms, runtime inflation -0.932803, mean novelty 0.837837, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

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TSP Adaptive Heuristic Portfolio Suite Review

Primary Endpoint: Held-Out TSPLIB Gap

- The best held-out TSPLIB gap (0.07226) is achieved by both:
- **phase8_best_single_fixed_heuristic** (farthest_insertion_two_opt)
- **phase8_supervised_ml_selector**
- The **oracle_selector** also shows this gap (0.07226) but is noted as an upper bound, not deployable.
- Other adaptive controllers have higher TSPLIB gaps (~0.20+), indicating no improvement over the best fixed heuristic in raw gap terms.

Selector Regret vs. Oracle Portfolio

- **phase8_best_single_fixed_heuristic** has zero selector regret (by definition).
- **phase8_supervised_ml_selector** matches zero selector regret, demonstrating near-oracle performance without runtime inflation.
- Adaptive controllers exhibit non-zero selector regret (0.13 - 0.14), indicating incomplete oracle approximation.

Runtime-Adjusted Gap & Runtime Inflation

- The best fixed heuristic and supervised ML selector have runtime inflation near zero or slightly negative, confirming no runtime penalty.
- The **phase8_best_single_fixed_heuristic** and **supervised_ml_selector** runtime-adjusted gaps are identical (~0.07226), efficient.
- Adaptive controllers show higher runtime-adjusted gaps (~0.20), but have negative runtime inflation (beneficial runtime reduction), indicating more efficient runtime behavior but at cost of solution quality.
- The **LLM static selector** has low runtime (296 ms vs. ~1500 ms for fixed heuristics) but substantially worse gap (0.2307), indicating a quality-runtime tradeoff.

Pareto Efficiency

- Pareto-efficient conditions:

- **phase8_supervised_ml_selector**: low gap, no runtime inflation, zero regret.
- **phase8_llm_evolved_adaptive_controller**: improved runtime (negative inflation), reasonable gap, interpretable adaptive control.
- **phase8_full_solver_evolution**: lowest runtime (~102 ms) but higher gap (0.2134), high code novelty suggesting newly evolved heuristics.

Cross-Family Transfer Gap

- Transfer gaps are smallest for oracle (0.027) and best fixed (0.031), followed closely by supervised_ml_selector (0.031).
- Adaptive controllers increase transfer gaps (0.09 to 0.11), suggesting less robust cross-family generalization.
- Random portfolio and full solver demonstrate higher transfer gaps, less effective generalization.

Interpretability and Instance-Adaptive Control

- The **phase8_supervised_ml_selector** achieves near-oracle performance with no runtime inflation and zero selector regret, confirming it as an effective static instance-adaptive selector.
- The **phase8_llm_evolved_adaptive_controller** provides interpretable, bounded online adjustments, achieving runtime gains with moderate quality trade-offs — a promising adaptive control approach.
- Increased complexity and code novelty in fully evolved controllers suggest exploratory solutions but with diminished interpretability and no quality gains.

Summary and Recommendations

- The known heuristic **farthest_insertion_two_opt** remains the strongest baseline in gap performance.
- The supervised ML selector matches the oracle gap and selector regret without runtime cost, demonstrating effective and interpretable instance-adaptive static selection.
- Adaptive controllers offer valuable runtime reductions with bounded quality loss, offering a practical trade-off for runtime-sensitive deployments.
- Cross-family transfer is best preserved by static selectors close to the oracle; adaptive controllers slightly degrade transfer robustness.
- Given the risk of overclaiming, conclusions emphasize:
- The strong baseline of the best fixed heuristic.
- Successful hyper-heuristic control via supervised ML static selector approximating oracle performance.
- Adaptive controllers offering runtime-efficient, interpretable instance-adaptive scheduling, without new heuristic discovery.

Key conservative claim:

Interpretable static and adaptive hyper-heuristic controller designs closely approach the oracle portfolio performance on held-out TSPLIB instances, with negligible runtime inflation or beneficial runtime reductions, demonstrating effective instance-adaptive control over known TSP heuristics.